

Chapter 9: Earth & Environmental Science

Geology, Earth Systems, Ecosystems, Biogeochemical Cycles & Climate Science

9.1 Earth Systems & Plate Tectonics

Earth is divided into four major subsystems: lithosphere, hydrosphere, atmosphere, and biosphere. Plate tectonics explains continental drift, earthquake zones, and volcanic activity driven by thermal convection currents in the mantle.

9.2 Ecosystem Dynamics & Biogeochemical Cycles

Ecosystems involve biotic communities interacting with abiotic factors. Energy flows unidirectionally through trophic levels (producers to apex predators), while matter cycles continuously through biogeochemical pathways (carbon, nitrogen, phosphorus, and water cycles).

9.3 Climate Science & Sustainability

Climate science investigates long-term atmospheric patterns. Greenhouse gases (CO_2 , CH_4 , H_2O) trap outgoing terrestrial radiation, maintaining Earth's habitable temperature. Sustainable resource management aims to balance human industrial needs with ecological conservation.